

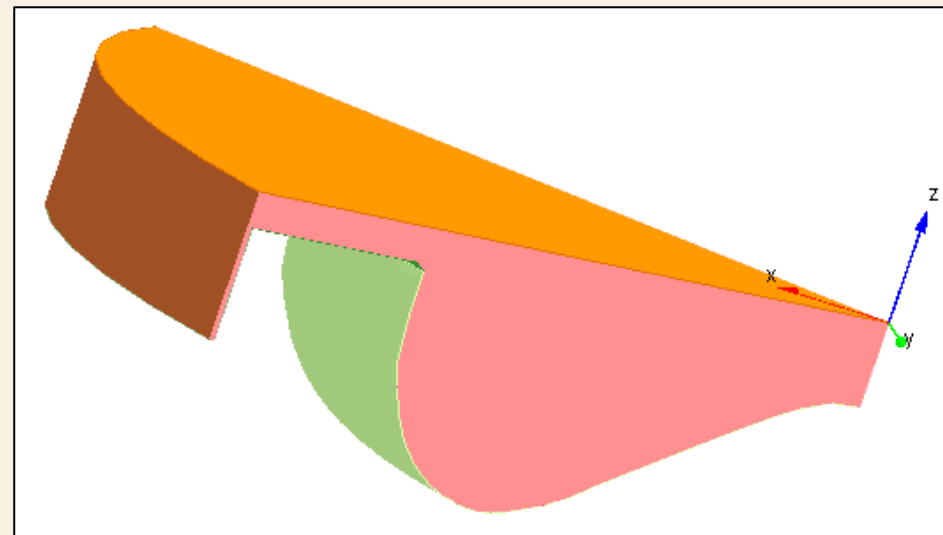


ANSYS FORTE

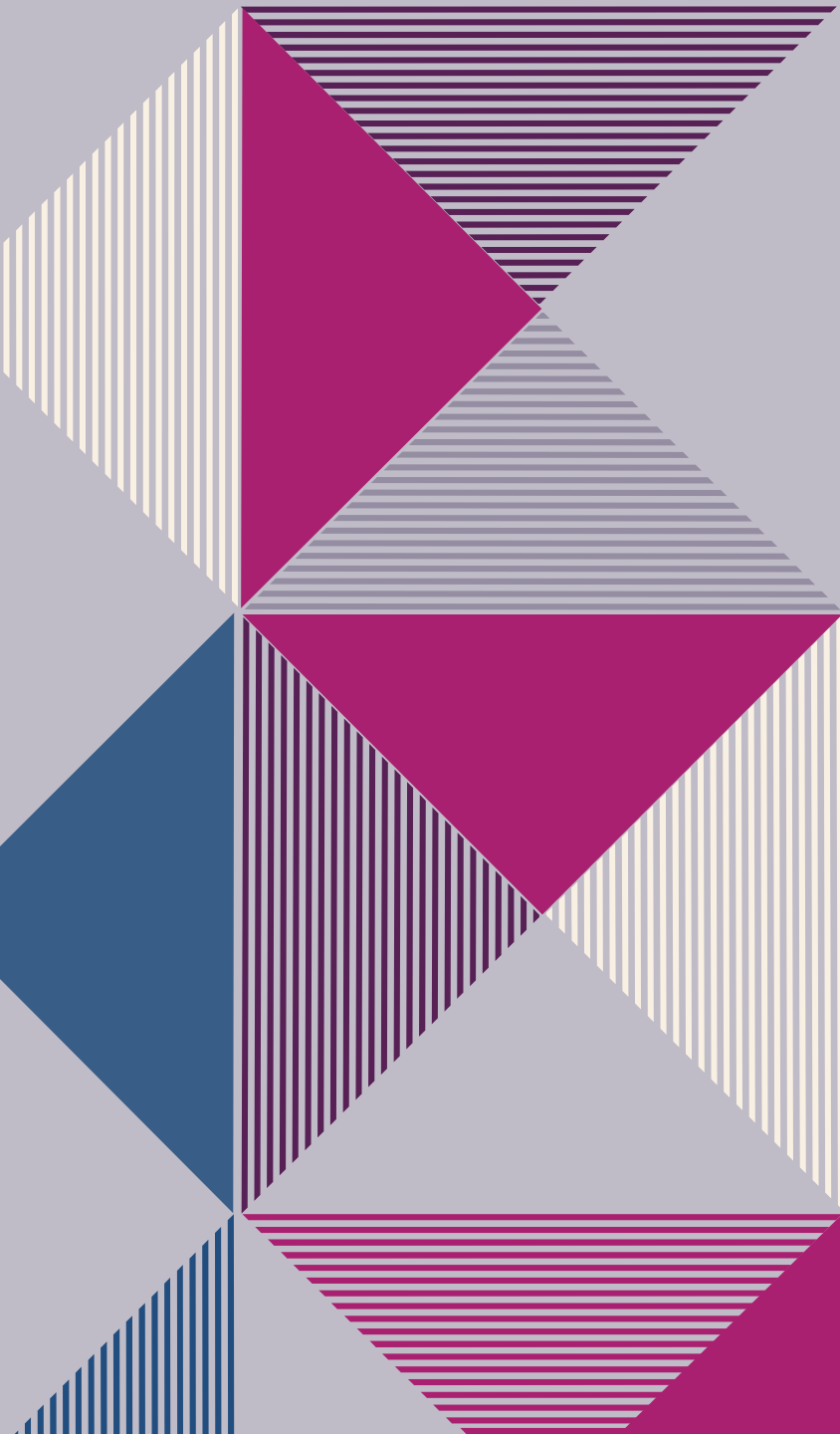
Simulating Dual Fuel Combustion

GEOMETRY

A picture of the geometry used is given below, it is divided into 7 sections



			periodic-A
			periodic-B
			head
			liner
			crevice
			piston-bowl
			piston-crown



MESHING

SURFACE REFINEMENT

- For the piston, head, and liner surfaces. 1 layer of cells of 1 mm
- For the crevice region. 1 layer of cells of 0.25 mm
- For the squish region. 2 layers of cells of 0.5 mm

POINT REFINEMENT

- cover the entire domain with 1 mm cells

CONE REFINEMENT

- uses cells of 1 mm

LINE REFINEMENT

- Along the sector mesh axis. Over a cylindrical region around the axis with a radius of 1 mm, 0.25 mm cells are used

SOLUTION ADAPTIVE

- Gradient of fuel vapor mass fraction, from near SOI -67 CA ATDC to slightly past expected ignition (+10 CA ATDC). 0.5 mm cells
- Gradient of velocity, from near SOI -67 CA ATDC to slightly past expected ignition (+10 CA ATDC). 0.5 mm cells
- Gradient of temperature, from near SOI -67 CA ATDC to 50 CA. 0.5mm cells

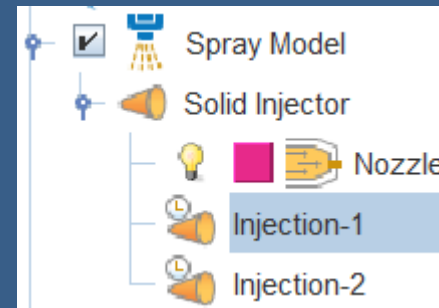
SET UP

CHEMISTRY

A custom chemistry file was used

SPRAY MODEL

- A Solid Cone injector is added
- Spray model: ON
- **SOLID CONE**
 - Droplet inflow temp: 360K
 - Gas jet model used
- **NOZZLE**
 - Nozzle Area = 0.000491 cm^2
- **INJECTION 1**
 - Duration: -67^0 to 5.46^0
 - Total Injected Mass: 0.015042g
- **INJECTION 2**
 - Duration: -2.73^0
 - Total Injected Mass: 0.007521g

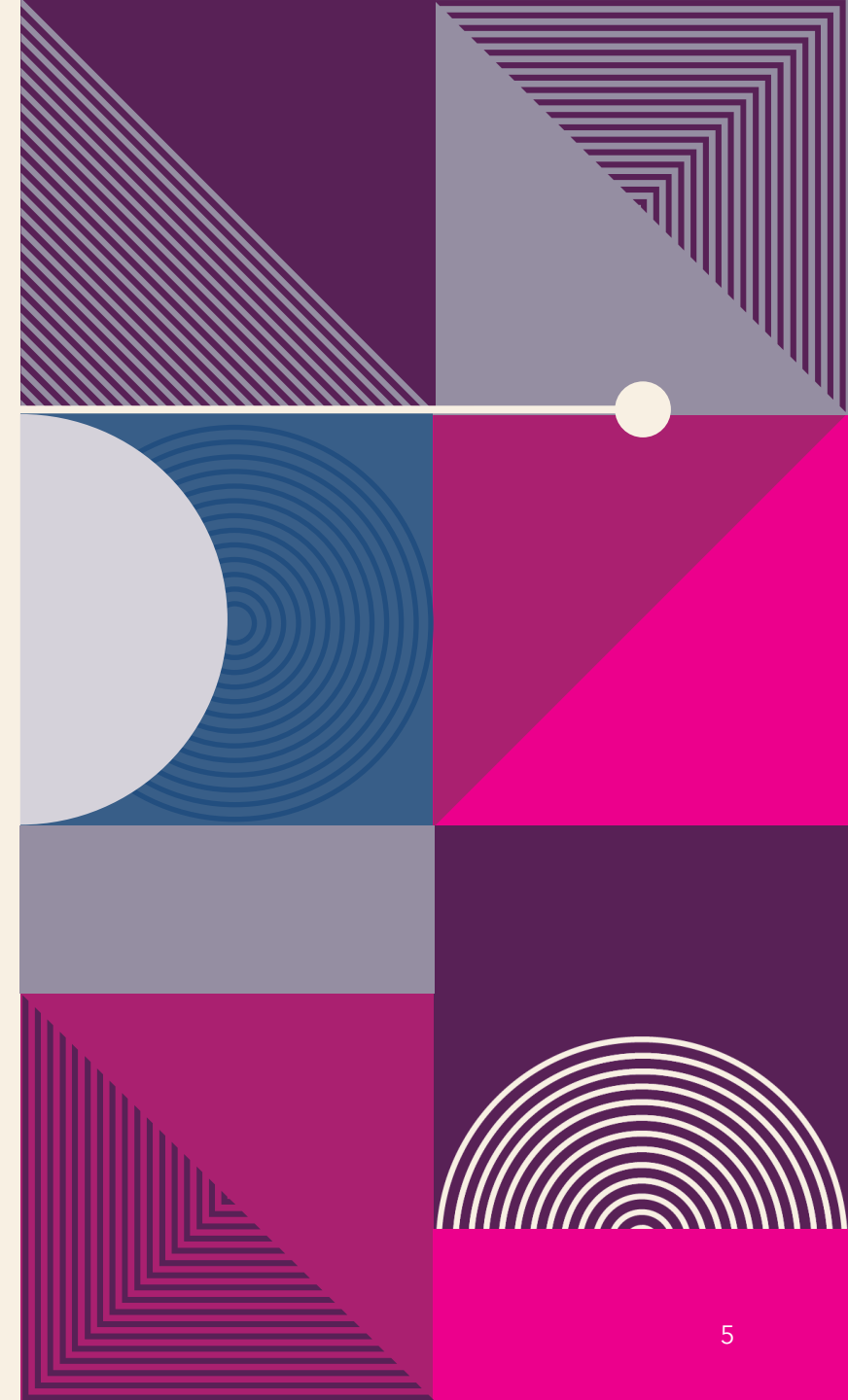


BOUNDARY CONDITIONS

WALL BOUNDARY

- The linear part of the geometry with $T = 430\text{K}$
- For piston wall temperature $T = 500\text{K}$; Wall motion = ON
- Head Temperature $T = 500\text{K}$
- Periodic boundary for sides
Periodic A and B Sector angle = 60°

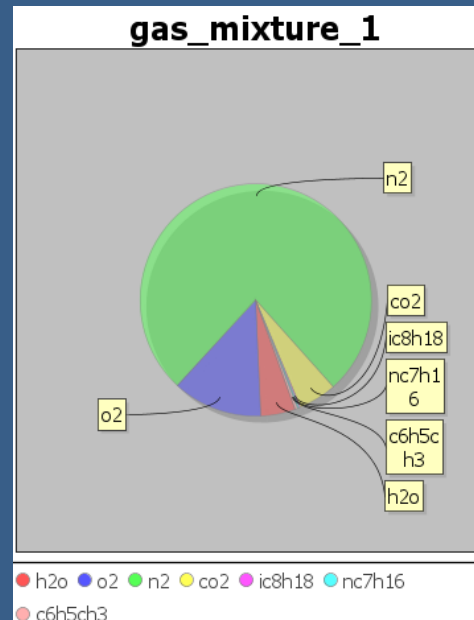
Stroke	16.51	cm	▼
Connecting Rod Length	26.16	cm	▼
Piston Offset	0.0	cm	▼



INITIAL CONDITIONS

- Composition -
- Temperature $T = 391\text{K}$
- Pressure = 3.34 bar
- Engine Swirl ratio = -0.7
- Initial swirl profile factor = 3.11

Composition		Mole Fraction
Species	Fraction	
h2o	0.049	
o2	1.25344E-1	
n2	7.65372E-1	
co2	0.053	
ic8h18	1.588E-3	
nc7h16	3.005E-3	
c6h5ch3	2.691E-3	



SIMULATION CONTROLS

SIMULATION END POINTS

- Simulation will be crank angle based
 - RPM: 1300
 - 4-stroke Cycle

CHEMISTRY SOLVER

- Dynamic cell clustering: ON
- Autoignition Induced flame propagation model
- Critical Temperature $T = 2000\text{K}$
- Critical Size coefficient: 1
- Chemistry activation is set to after fuel injection starts the threshold temp is $T = 600\text{K}$

TIME STEP

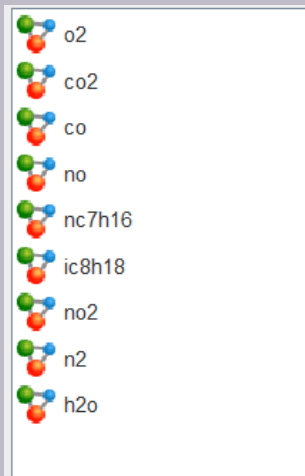
 **Max. Time Step Option** ▼

Max. Simulation Time Step ▼

OUTPUT CONTROLS

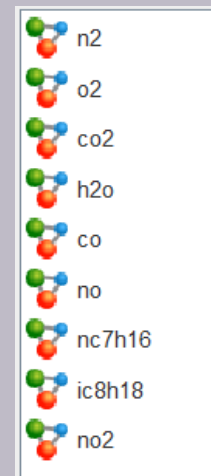
SPATIALLY RESOLVED

- Output Every :5.0⁰
- Spatially resolved species



SPATIALLY ARRANGED

- Output every 5 degree
- Spatially arranged species



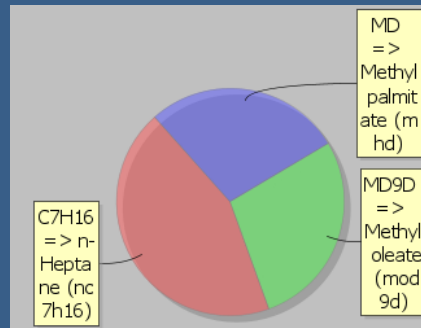
RESTART DATA

- Write restart file at the last step of the simulation

SET UP(FOR BIO-DIESEL)

CHEMISTRY

A Multifuel reaction Mechanism was used which was converted and processed to make it compatible to work with Ansys (it consisted of the chemistry, transport and thermos data)

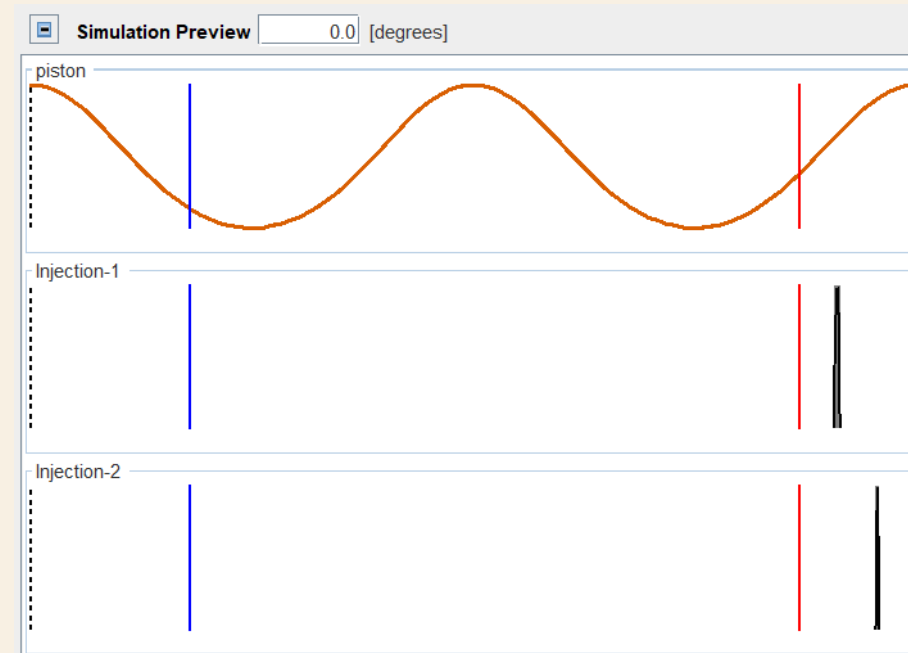
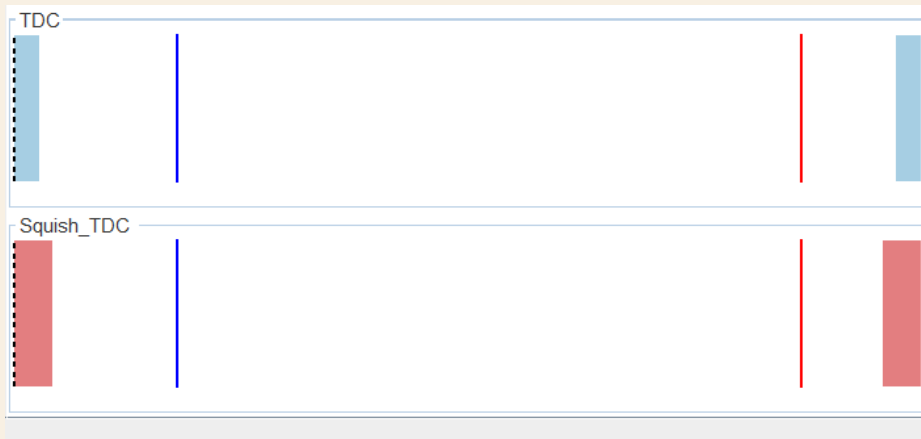


BIO-DIESEL

- To make a 100% Biodiesel composition consisting of mass fraction of 44% of C7H16(n-Heptane), 28% of MD(Methyl palmitate), and 28% of MD9D(Methyl oleate)
- for 20% biodiesel, these quantities were changed to 5.6, 5.6 and 88.8 percent respectively for MD9D, MD and C7H16
- For 50% Biodiesel composition consisting of mass fraction of 72% of C7H16, 14% of MD and MD9D

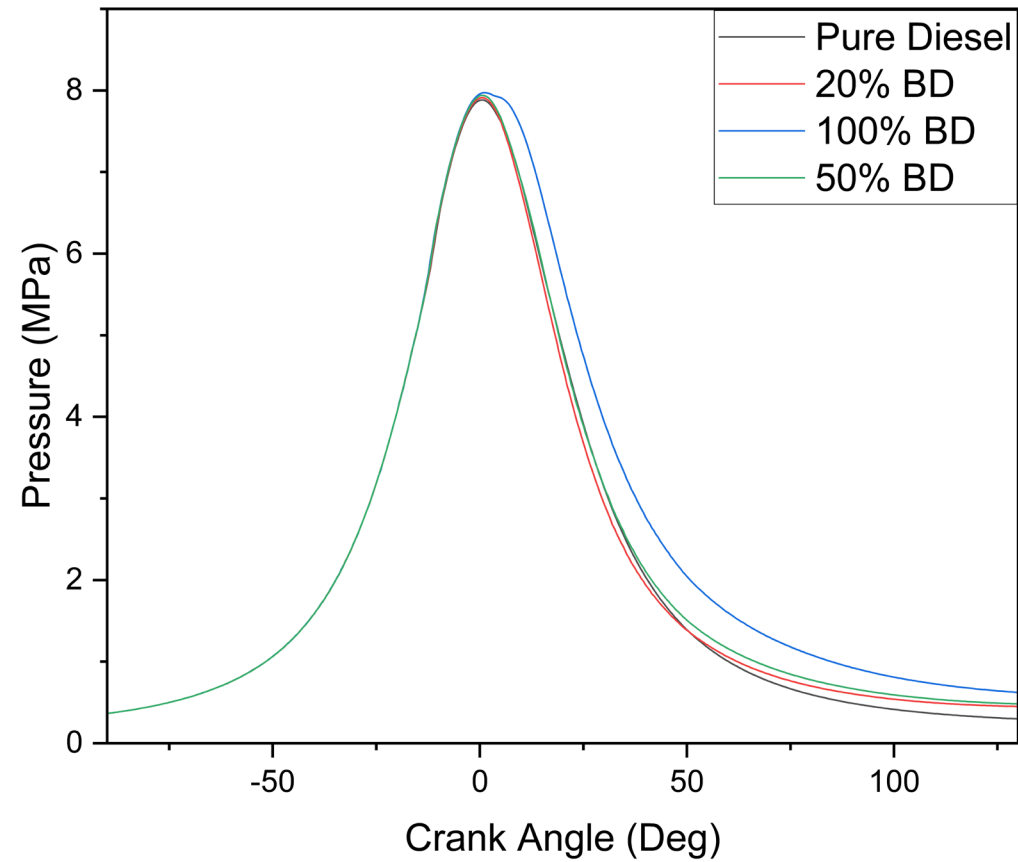
RESULTS

A Total of 4 run were done with pure diesel(black), 20% Biodiesel(Red), 100% Biodiesel(Blue), and 50% Biodiesel(Green) parameters like Max Temperature, Pressure, CO, NO2 and Unburned Hydrocarbons are plotted

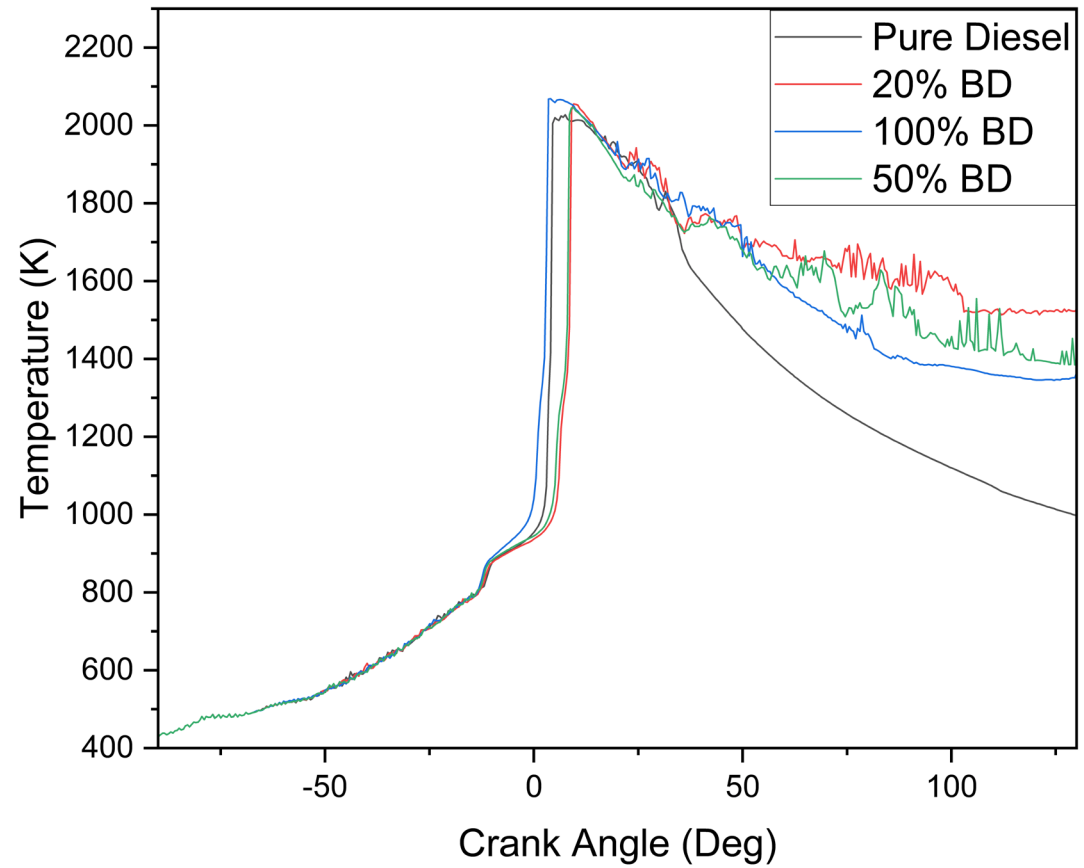


Biodiesel blends and pure biodiesel seem to produce fewer emissions as compared to Purediesel as observed in the graph

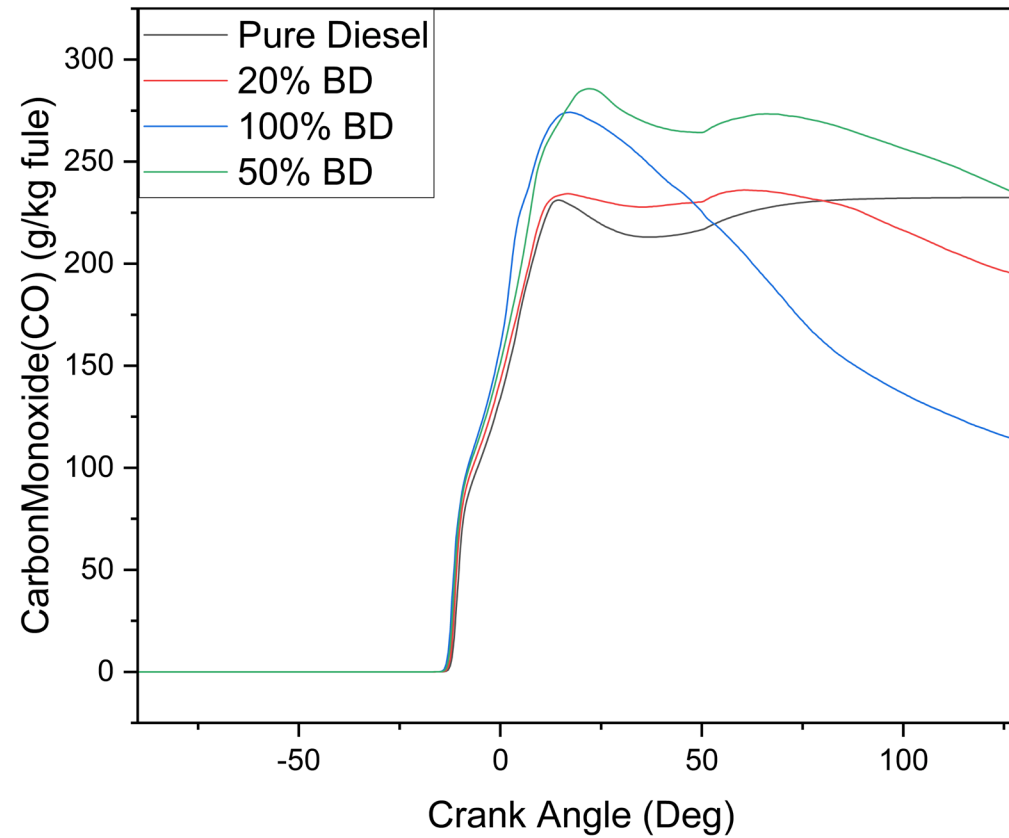
PRESSURE VS CRANK ANGLE



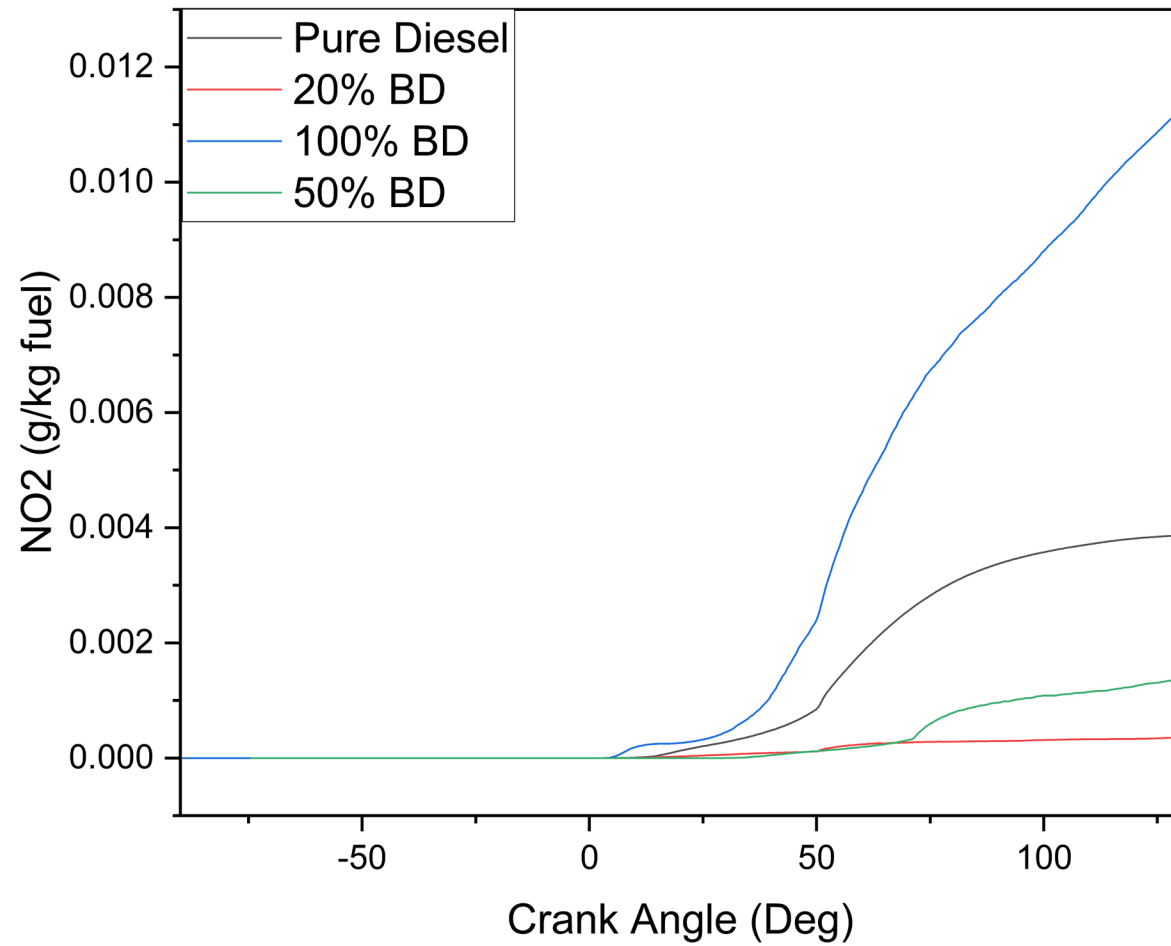
MAX TEMPERATURE GRAPH



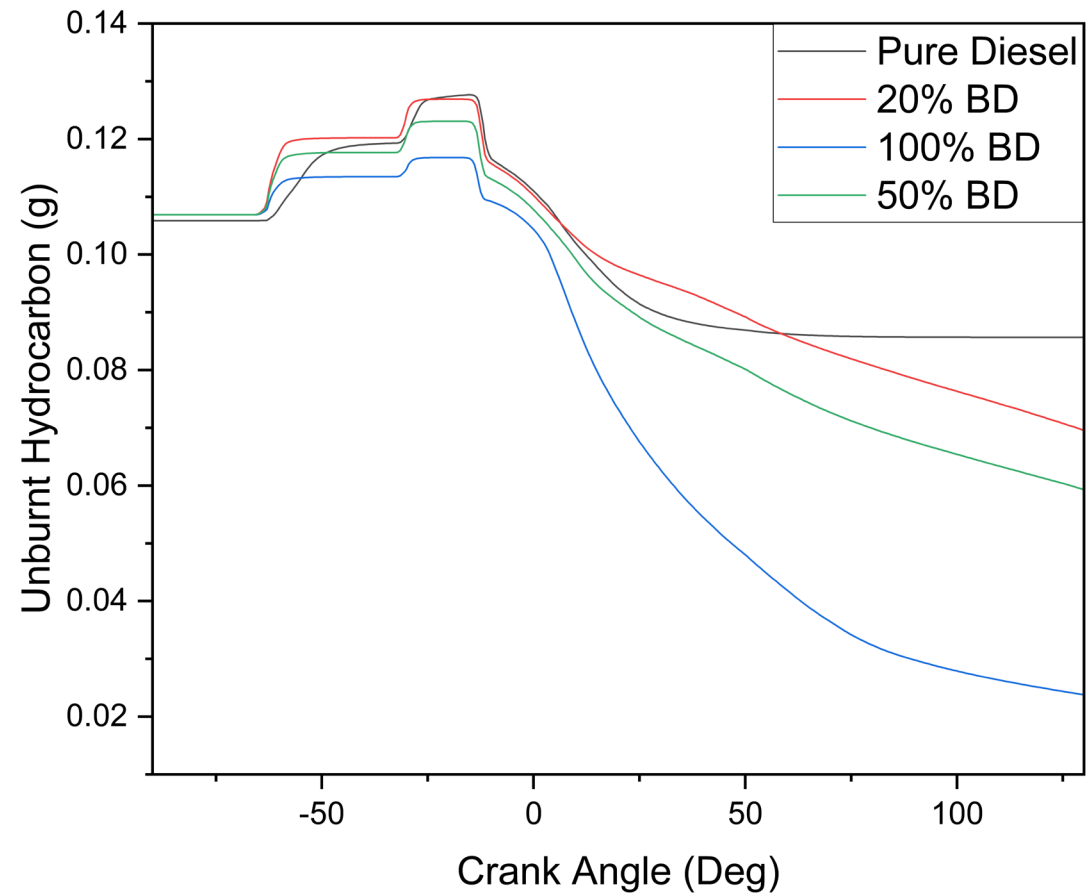
CO (CARBON MONOXIDE)



NO₂



UNBURNT HYDROCARBON



An abstract geometric design on the left side of the slide. It features a dark blue background with various geometric shapes and patterns. A white circle is positioned near the top left. Below it, a light blue semi-circle is visible. To the right of the semi-circle, there is a pink triangle with diagonal lines. Further down, there is a pink square with a pattern of concentric lines. At the bottom, there is a pink triangle with a pattern of concentric lines. The overall design is modern and minimalist.

THANK YOU